NEW YORK CITY UNDER THE SPOTLIGHT

Exploring the 2021 NYPD Arrest Dataset

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Executive Summary

- Arrest activity is highest in Brooklyn, followed by Manhattan and the Bronx
- Adults aged 25–44 make up the largest share of arrests
- Most arrests involve male individuals, with a large gap compared to females
- Arrest data also shows differences across racial groups, though this may reflect policing patterns
- Assault and theft-related offenses are the most common

Recommendation: Focus safety resources and infrastructure in high-activity areas, especially where arrest patterns are most concentrated

The Problem

“Murders and shootings in New York **spiked in 2020** during the onset of the Covid-19 pandemic. That year, **shootings nearly doubled and murders jumped more than 40 percent** compared to 2019. Murders climbed again, by 4 percent, between 2020 and 2021.”

(Brennan Center for Justice, 2025)

Dataset Overview

2021 NYPD Arrest Record

155,404 Total Arrests

All **5 Boroughs | 77 Precincts**

Felonies, Misdemeanors & Violations

Dataset sourced from NYC Open Data (2021 NYPD Arrest Records)

KEY VARIABLES

Demographics: Race, age, and sex

Specific Offense Descriptions

Geospatial Data: Latitude & Longitude

ACTIONABLE INSIGHTS

Identify "High Incident" zones

Inform Infrastructure Placement

Equity Analysis

Key Questions

1. Where are arrests most concentrated across NYC?
2. What patterns exist across age, gender, and offense type?
3. How can this data help guide safer infrastructure placement?

Our Goals

- Identify key patterns in arrests (age groups, offense types, and locations)
- Understand which areas show higher arrest activity across NYC
- Help inform where police resources can be reinforced in a fair and effective way
- Pinpoint where safety improvements like streetlights or emergency infrastructure could be added
- Provide data-driven recommendations to improve public safety for NYC officials and stakeholders

Stakeholders: NYC officials, law enforcement, and the public. All working toward improving safety in the city.

Methodology

Disclaimers – Limitations of Arrest Data:

Understanding Reporting Bias

Arrest ≠ crime

dataset captures enforcement activity, not actual crime occurrence.

Precinct coverage bias

data reflects where police are, not necessarily where crime is

Stop-and-frisk legacy

certain communities are over-policed, inflating arrest counts

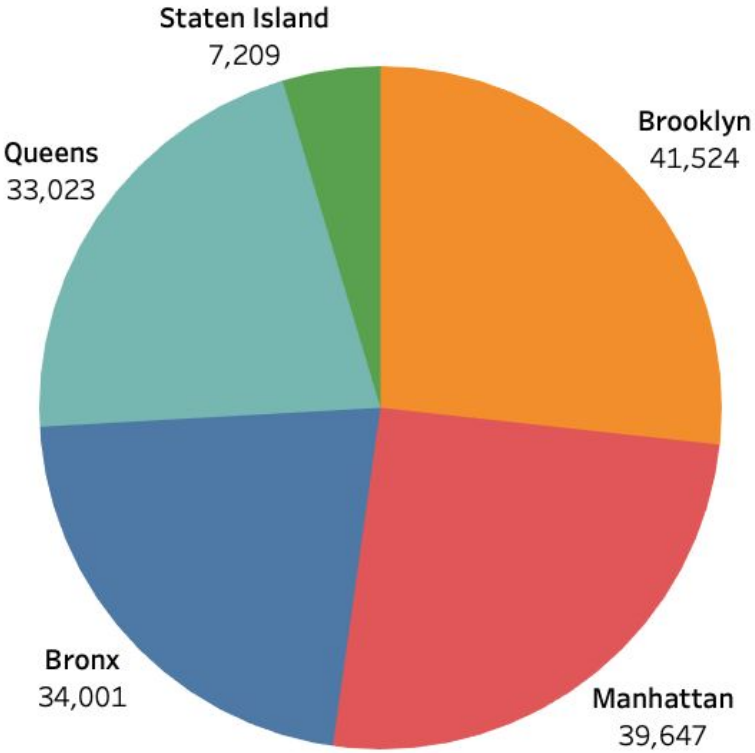
Single year snapshot

2021 was atypical post-COVID

Our MAJOR

FINDINGS

Arrest By Borough

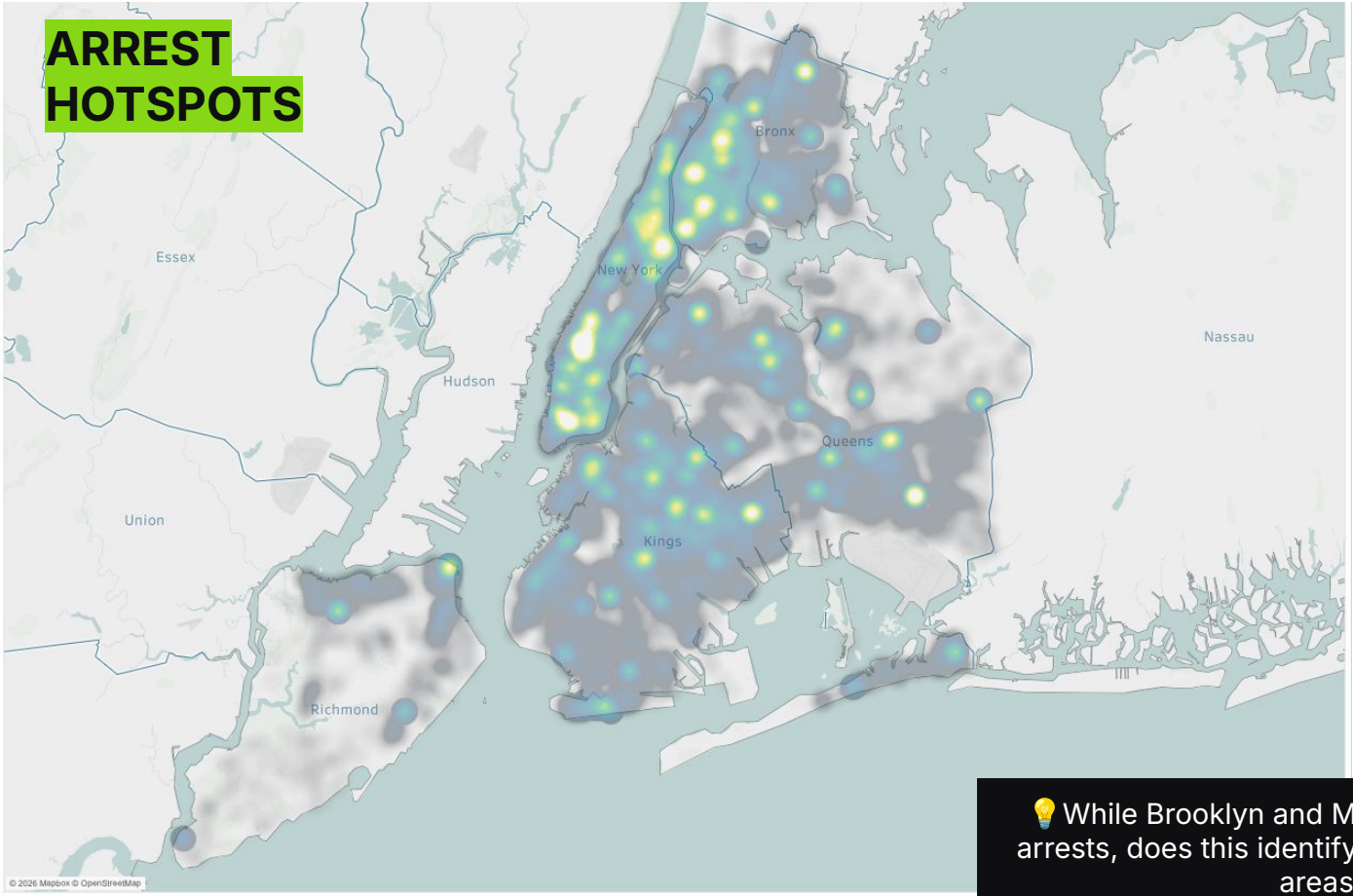


Key insights from Arrest by Borough

- **Brooklyn** has the **highest number of arrests**, making it the most active borough in the dataset
- **Manhattan** and the **Bronx** follow closely behind, showing consistently high levels of activity
- **Queens** has slightly fewer arrests but is still a major contributor overall
- **Staten Island** has the **lowest number of arrests** by a large margin
- **Overall**, arrest activity is concentrated in a few boroughs, which can help guide where safety resources and infrastructure should be prioritized

While the previous chart shows overall distribution by borough, this map highlights where arrests are concentrated within those areas.

ARREST HOTSPOTS



Arrest Counts by Borough

BROOKLYN: 41,524

MANHATTAN: 39,647

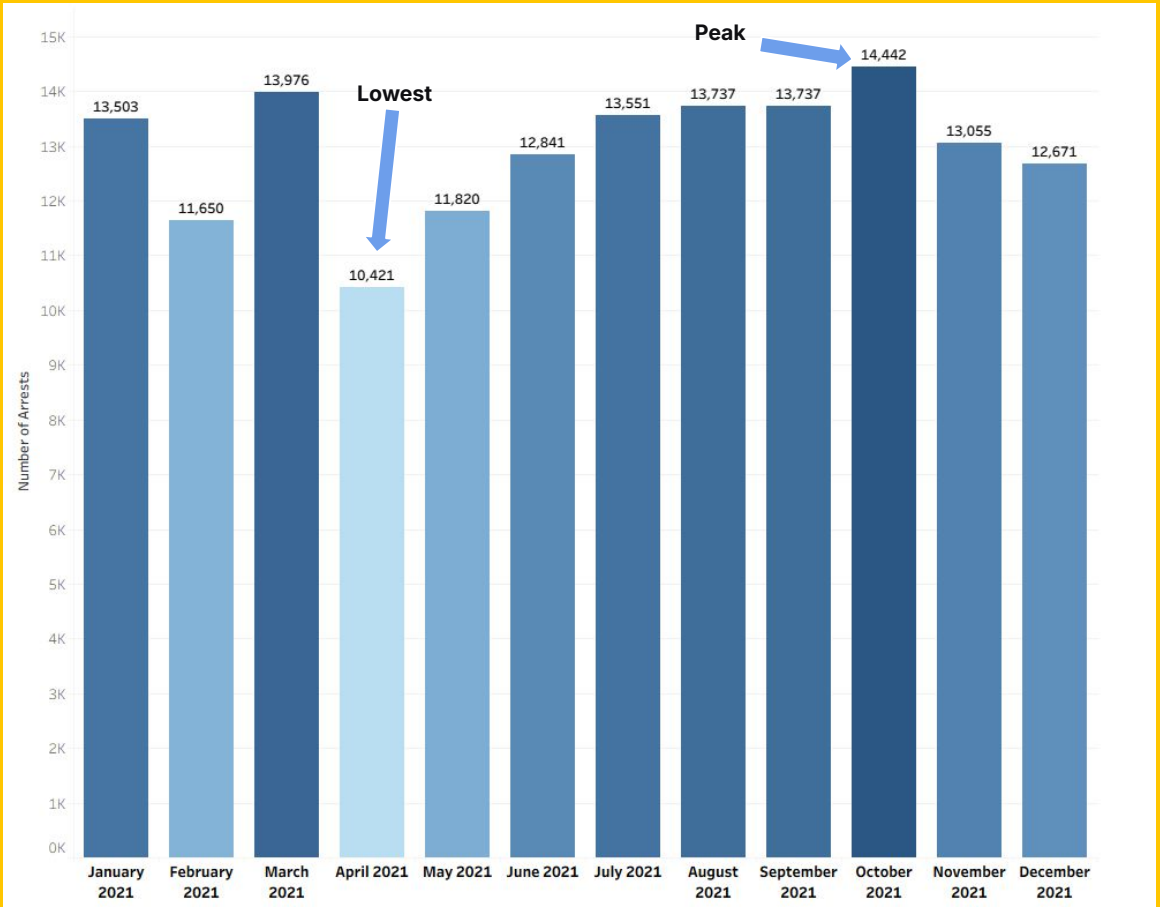
BRONX: 34,001

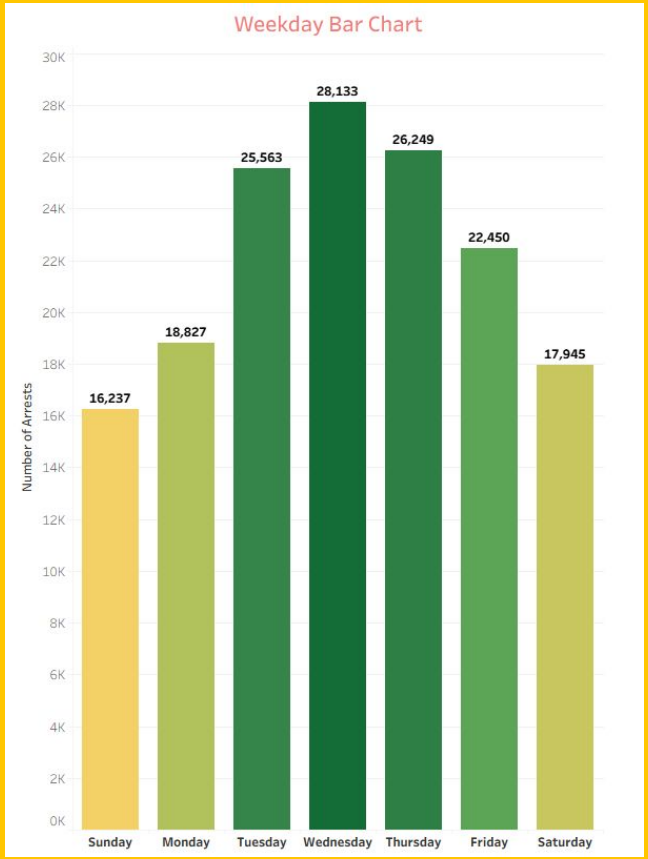
QUEENS: 33,023

STATEN ISLAND: 7,209

💡While Brooklyn and Manhattan show high arrests, does this identify the best investment areas?

Monthly and Weekday Arrest Trends

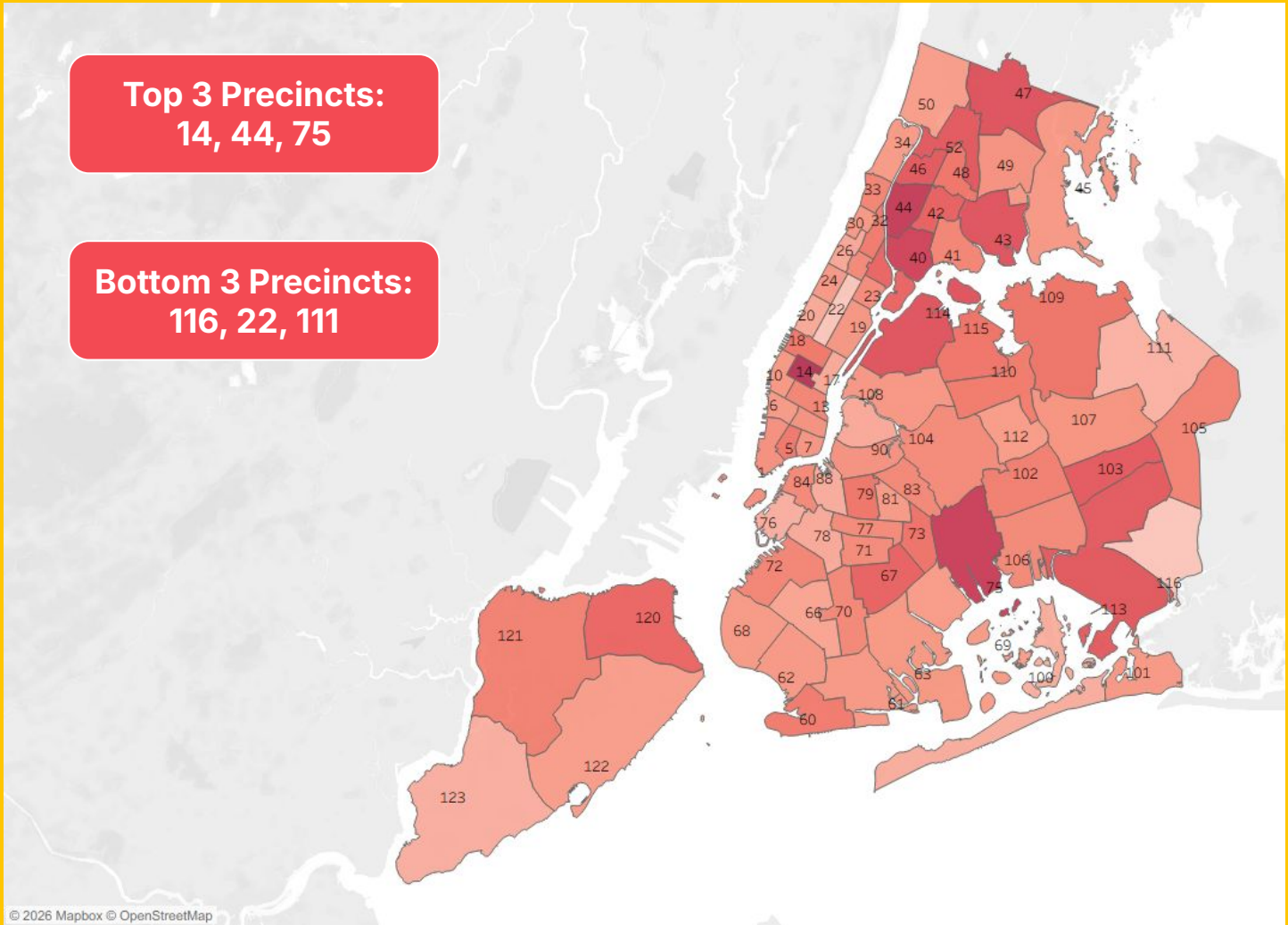




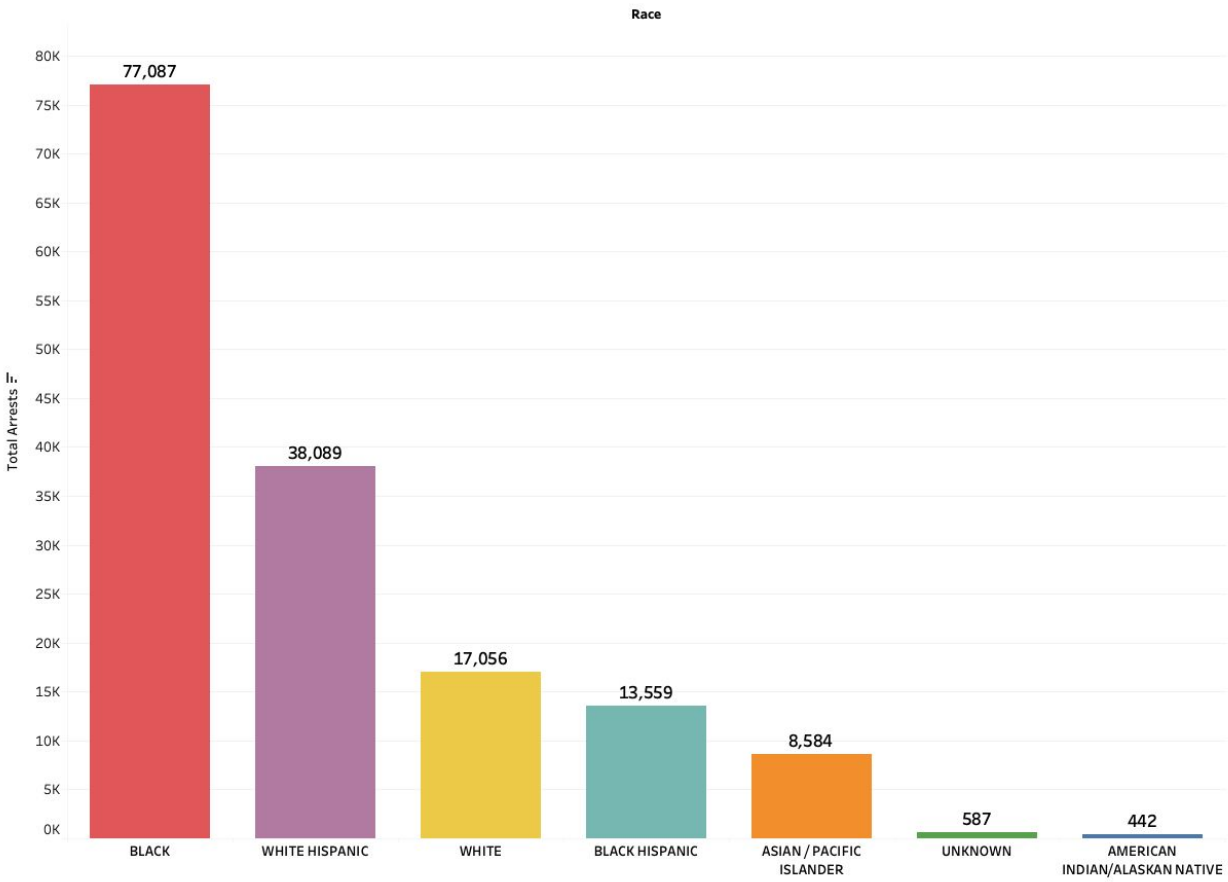
Heat Map of Precincts

Top 3 Precincts:
14, 44, 75

Bottom 3 Precincts:
116, 22, 111



Arrest by Race



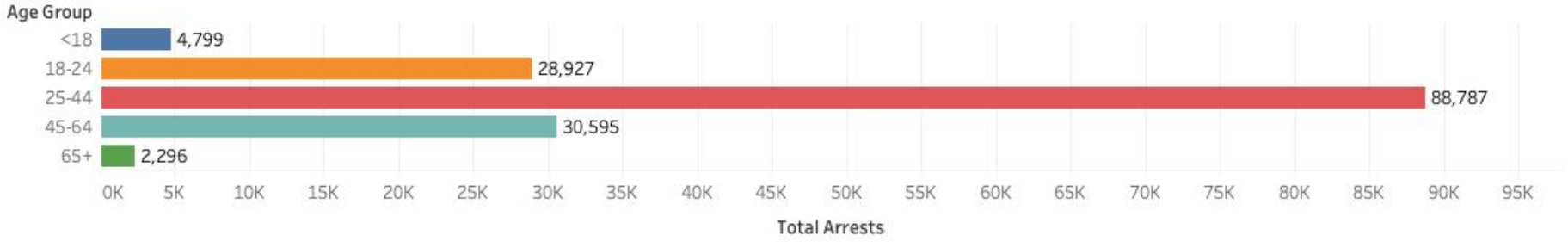
Key Insights from Arrest by Race

Major Arrest Demographics:

- **Black Individuals:** Constitute the largest demographic with 77,087 incidents, representing 49.6% of the 155,404 total arrests documented.
- **Hispanic Breakdown:** Collectively accounts for approximately 33.2% of arrests, segmented into White Hispanic individuals at 24.5% (38,089) and Black Hispanic individuals at 8.7% (13,559).
- **Distribution Among Other Races:**
  - White individuals account for 11.0% (17,056) of total incidents.
  - Asian / Pacific Islanders comprise 5.5% (8,584) of arrests.
  - American Indian/Alaskan Native and Unknown categories combined total less than 1%.

These statistics indicate significant demographic disparities and suggest potential variations in enforcement intensity across racial groups. Overall, the data emphasizes substantial demographic imbalances in arrest patterns, necessitating a more thorough examination of how enforcement is applied across different racial populations.

Arrest by Age Group

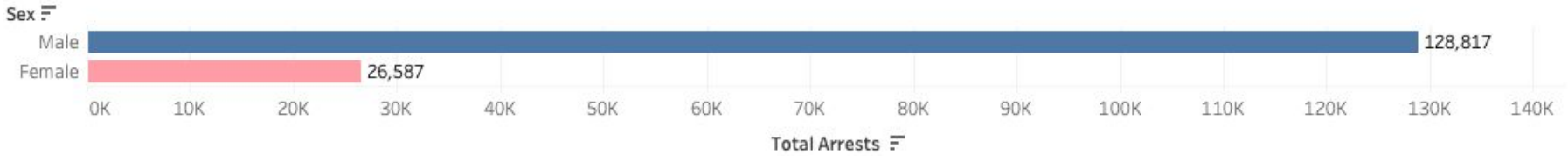


Key Insights from Age group

- Adults aged **25–44** make up the **largest** share of arrests by a wide margin **(~88.8K)**
- Ages **18–24** also contribute a significant number of arrests **(~28.9K)**
- Ages **45–64** show moderate activity **(~30.6K)**, indicating involvement across older working-age groups
- Juvenile arrests **(<18)** are relatively low **(~4.8K)**
- Seniors **(65+)** have the lowest number of arrests **(~2.3K)**
- **Overall**, arrests are **heavily concentrated** in the **25–44** age group

This suggests prevention and outreach efforts should focus more on working-age adults, where most arrest activity is concentrated.

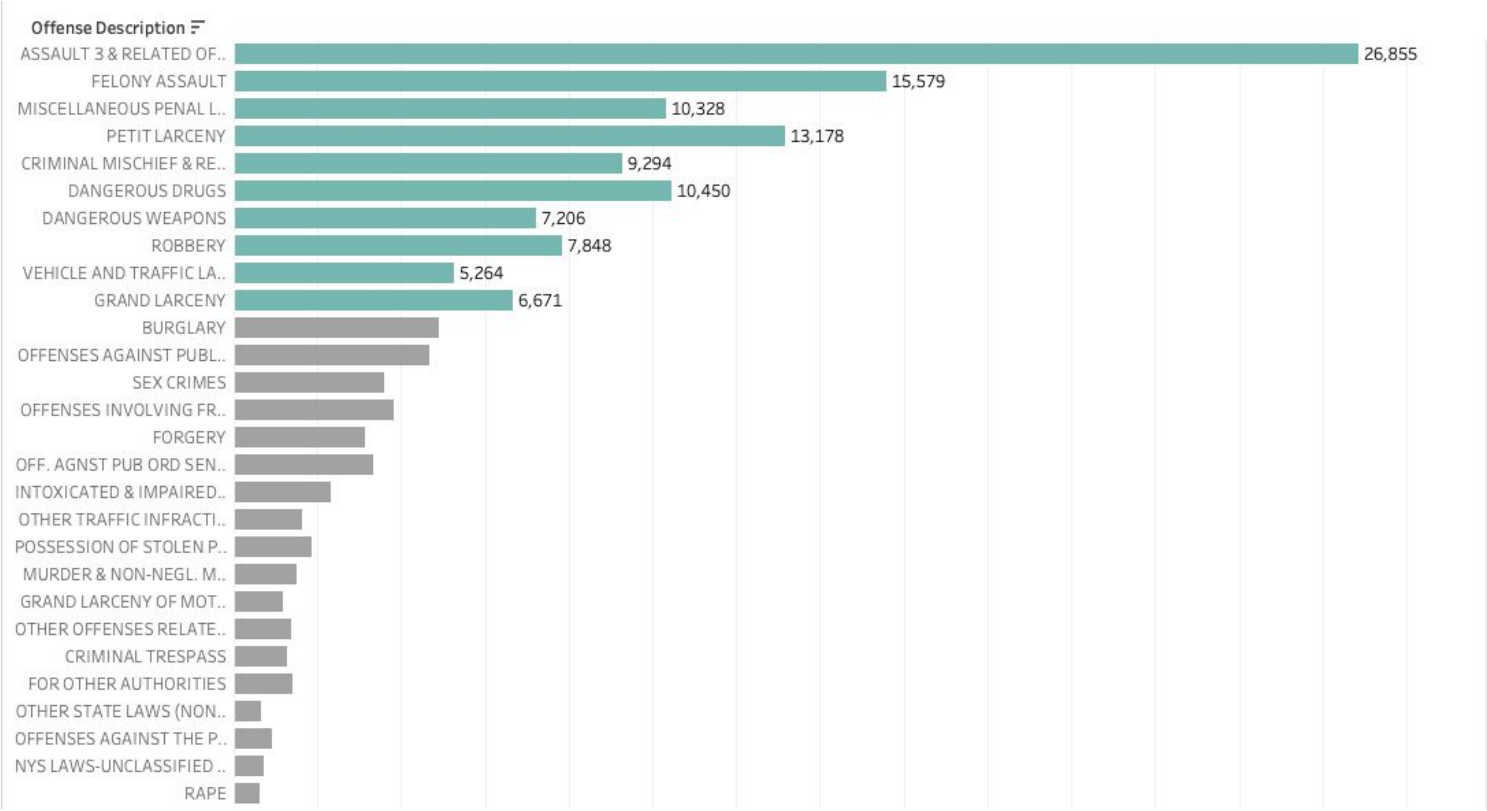
Arrest by Sex



Key Insights from Arrest by Sex

- **Male** individuals account for the **majority** of arrests, with approximately **128.8K** cases
- **Female** arrests are **much lower**, at around **26.6K**
- **Overall**, **male** arrests occur at **nearly five times the rate of female arrests**
- This pattern is consistent with trends seen in most criminal justice data
- This **suggests** prevention and intervention efforts may need to **focus more on the groups most involved**

Top 10 Arrest Offenses



Top 10 Arrest Offenses: Key Insights

Arrests are heavily concentrated in assault and theft-related offenses, suggesting these areas need the most attention

Key findings:

- **Assault 3 & Related Offenses** is the most common category by a wide margin **(~26.9K)**, followed by **Felony Assault (~15.6K)**
- **Property crimes** like Petit Larceny **(~13.2K)** and Grand Larceny also make up a large portion of arrests
- **Drug-related offenses (Dangerous Drugs, ~10.4K)** are another major contributor
- Mid-level categories like **Miscellaneous (~9.3K)** and **Criminal Mischief (~7.8K)** remain consistent
- Other offenses like **Robbery (~6.7K)** and **Dangerous Weapons (~5.3K)** appear **less frequently** but are still notable

This shows that focusing on a few high-frequency offense types could have the biggest impact on overall public safety.

Current NYC Efforts to Address Crime & Equity

- Uses data-driven policing (CompStat) to track crime patterns and trends
- Implements performance management systems to reduce crime and improve outcomes
- Deploys officers to high-crime areas based on real-time data insights
- Provides public access to crime data to increase transparency
- Maintains accountability through data monitoring and reporting systems

Conclusion & Recommendations

Conclusion:

- Arrest patterns vary across boroughs, age groups, and offense types
- Arrests are concentrated in specific areas like Brooklyn, Manhattan, and the Bronx
- Demographic disparities suggest differences in enforcement across groups
- Arrest data reflects policing activity, not just actual crime

Recommendations:

- Focus safety resources like lighting, emergency phones, and patrols in high-activity areas like Manhattan and the Bronx
- Combine data with community context to reduce potential bias in enforcement decisions
- Prioritize prevention efforts for the groups most impacted
- Continue improving transparency and accountability through data monitoring

Call to Action:

Use these insights alongside current NYC strategies to make more targeted, equitable decisions on where safety resources and infrastructure should be placed across NYC

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Thank You for Listening

Q&A